
Open Source
Sustainability
Richard Littauer - http://richard.social
Rimu https://www.inaturalist.org/observations/198921399

Who is this Richard Littauer, anyway

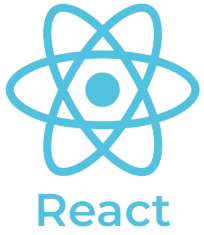
- PhD student in CompSci focusing on building better tooling for community science data of birds
- Organizer of **CURIOSS**
- Organizer of **SustainOSS**
- Organizer of **Open Sustain Technology**
- Former Executive Director of Gnome
- Ashthagn a kazudi

Source not given

Open source:













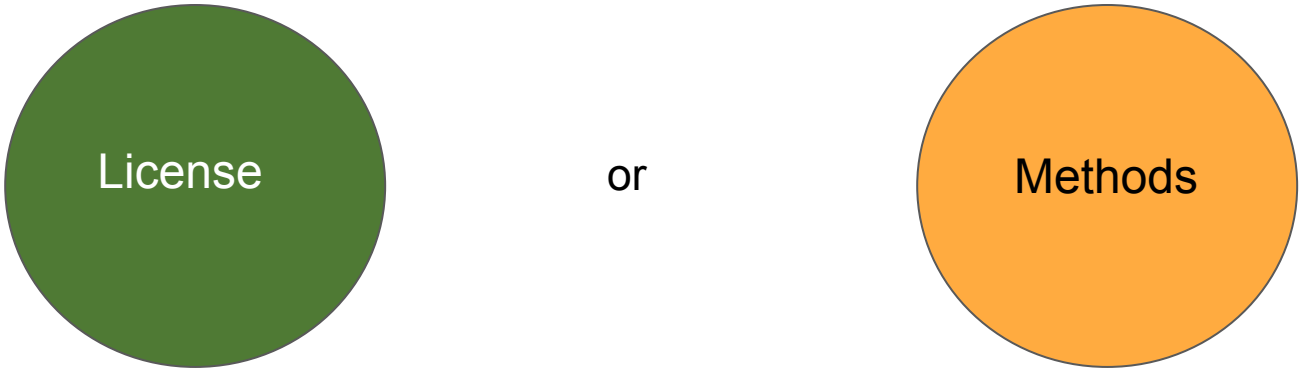




left-pad

```
1  module.exports = leftpad;
2
3  function leftpad (str, len, ch) {
4    str = String(str);
5
6    var i = -1;
7
8    ch || (ch = ' ');
9    len = len - str.length;
10
11
12   while (++i < len) {
13     str = ch + str;
14   }
15
16   return str;
17 }
```

Open source:



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How much open source?

GitHub

150+ million	4+ million	420+ million	90%
Developers	Organizations	Repositories	Fortune 100

How much open source?



ecosyste.ms

2024 WORKING PAPER HBS WORKING PAPER SERIES

The Value of Open Source Software

By: Manuel Hoffmann, Frank Nagle and Yanuo Zhou

Format: Print | **Language:** English | **Pages:** 42

Email Print Share

ABSTRACT

The value of a non-pecuniary (free) product is inherently difficult to assess. A pervasive example is open source software (OSS), a global public good that plays a vital role in the economy and is foundational for most technology we use today. However, it is difficult to measure the value of OSS due to its non-pecuniary nature and lack of centralized usage tracking. Therefore, OSS remains largely unaccounted for in economic measures. Although prior studies have estimated the supply-side costs to recreate this software, a lack of data has hampered estimating the much larger demand-side (usage) value created by OSS. Therefore, ge unique global da Ve first estimate th We then calcula software and wo f widely-used OSS nd that firms would The top six program Further, 96% of the demand-side value is created by only 5% of OSS developers.

We estimate the supply-side value of widely-used OSS is $4.15 billion, but that the demand-side value is much larger at $8.8 trillion.

ABOUT THE AUTHOR



Frank Nagle
Strategy
→ More Publications

MORE FROM THE AUTHORS

JANUARY 2025 FACULTY RESEARCH

AGENTS.INC: PATHWAYS TO GROWTH AT AN AI STARTUP

By: Frank Nagle and Susan Pinckney

NOVEMBER 2024 FACULTY RESEARCH

FEDEX CYBERATTACK (B): REFLECTIONS AND LESSONS

By: Hise Gibson, Frank Nagle, Alicia Dadlani and Martha Hostetter

NOVEMBER 2024 FACULTY RESEARCH

FEDEX CYBERATTACK (A): NAVIGATING THE NOTPETYA STORM

By: Hise Gibson, Frank Nagle, Alicia Dadlani and Martha Hostetter

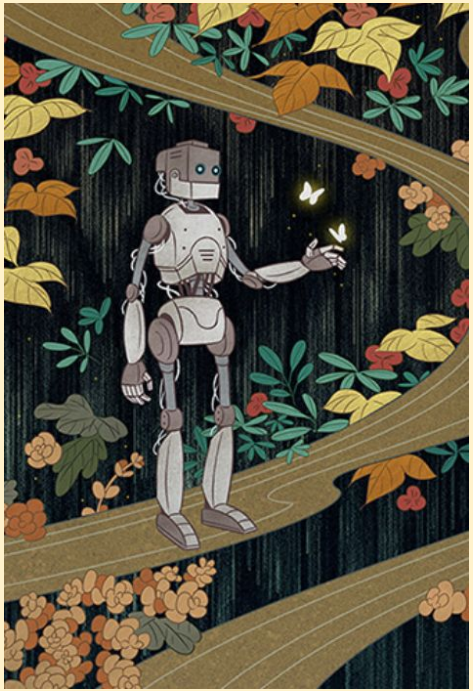
Why would you produce Open Source?

- Collaborate with others
- More eyes = less bugs
- Improve your CV
- Status and clout
- More adoption
- "Feels good"
- Security (never roll your own)

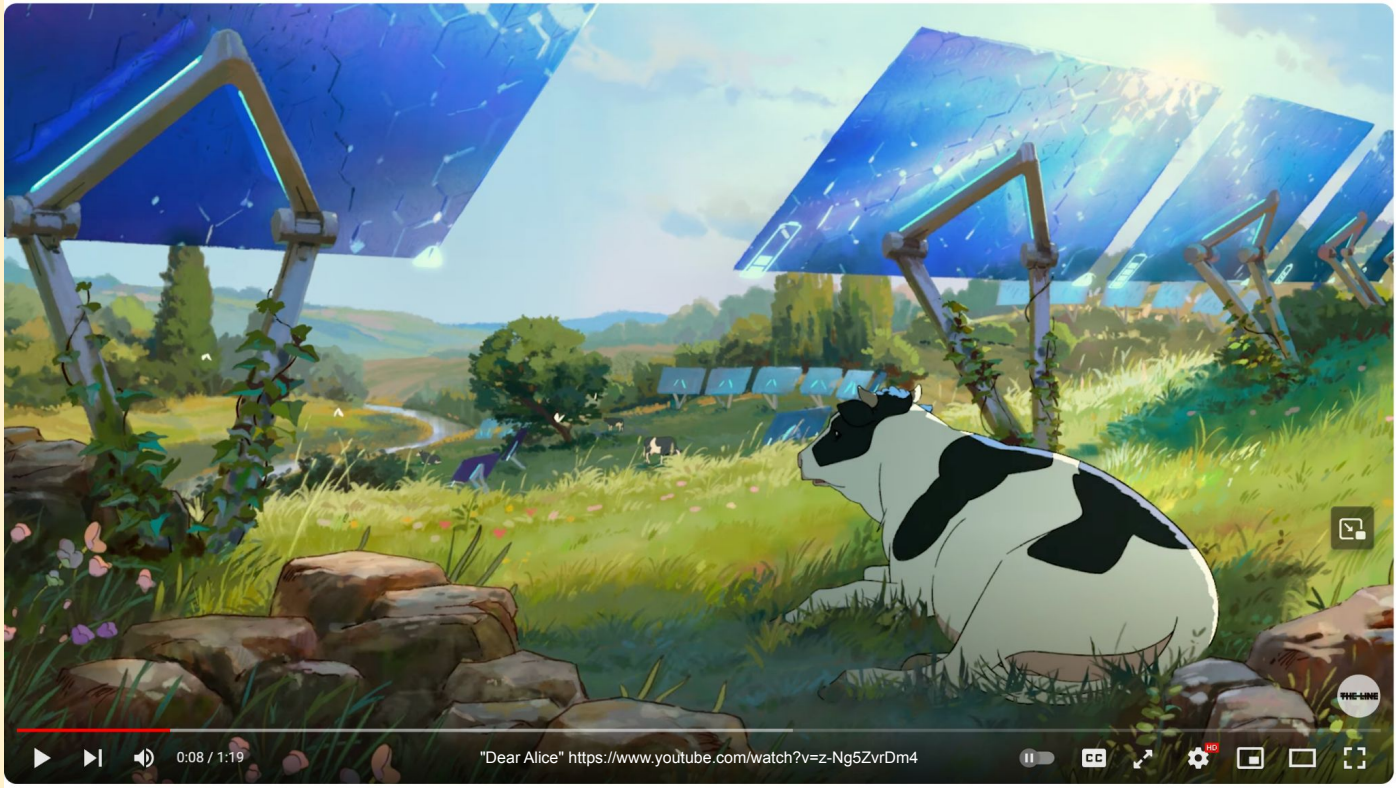
Why would you **NOT** use Open Source?

- Extractive use cases
- Poor expectation setting for users
- Too transparent (medical, indigenous rights)
- Too much work building community
- Business logic could be proprietary
- Security (never roll your own)

What is Sustainability?



https://www.behance.net/gallery/127556565/Book-Cover-A-MONK-AND-ROBOT-BOOK



Where sustainability applies in FOSS:

- Projects
- Developers
- Communities
- Businesses
- Academia
- Ecosystems
- Environment


Ramps https://www.inaturalist.org/observations/208351861

Sustain

Sustain: FOSS Projects

- Growth
- Diversity
- Funding

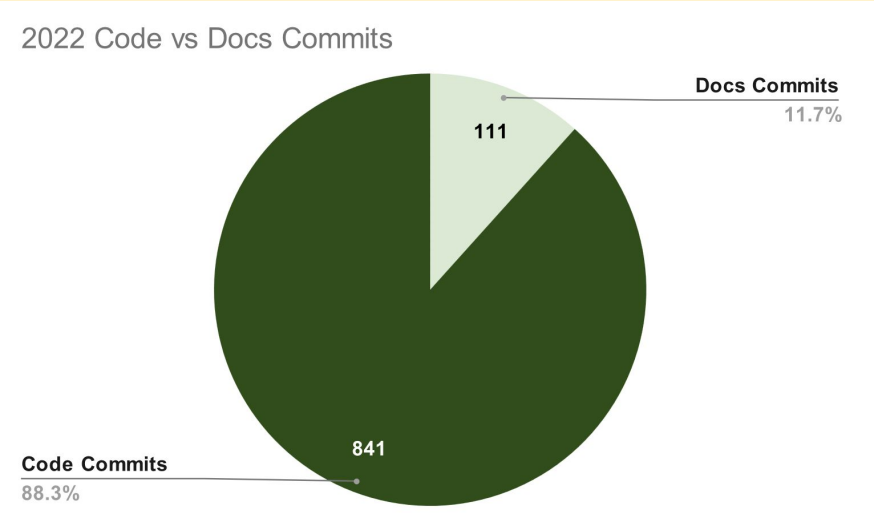
Sustain: FOSS Projects

- Growth
  - More in than out
  - Pathways to maintainership
  - Product management and design planning
  - Goal: capturing market share of attention



Sustain: FOSS Projects

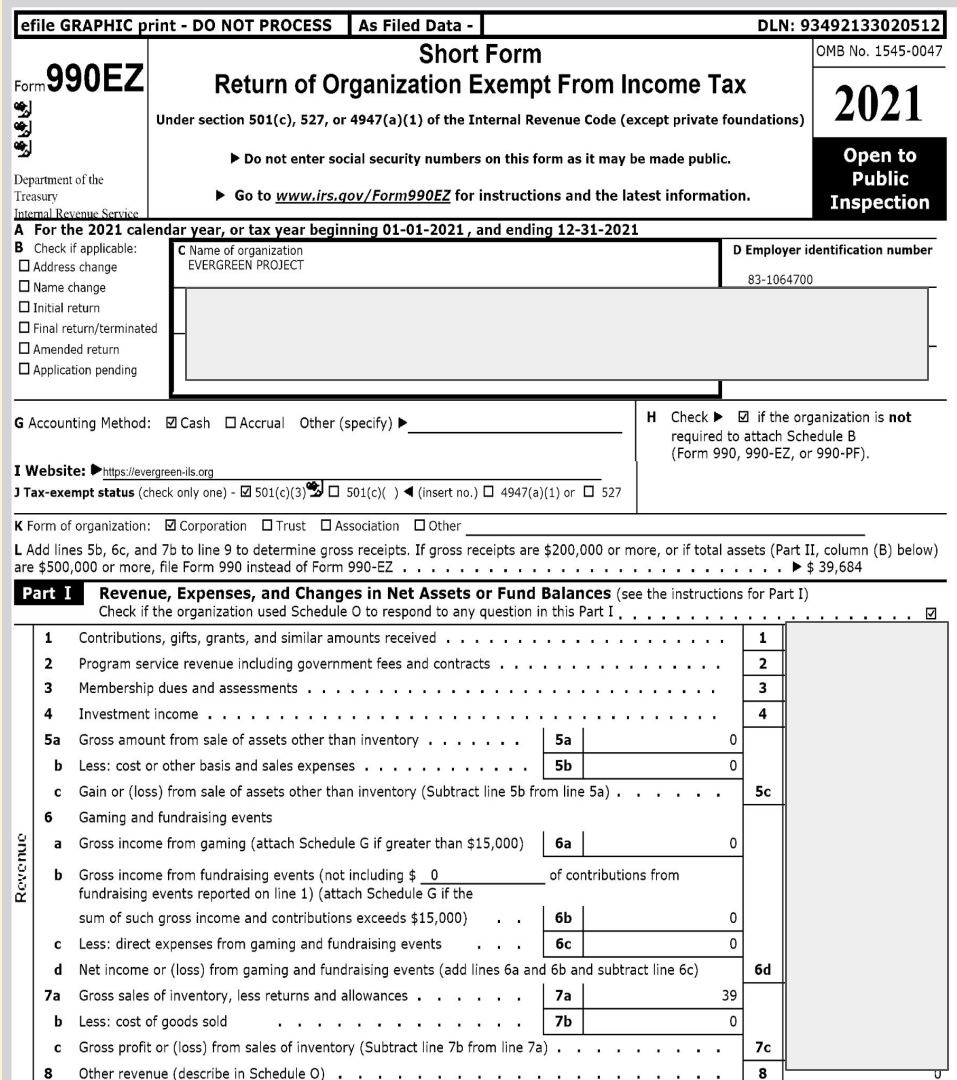
- Diversity
  - Welcoming community spaces
  - CoCs
  - Documentation
  - Moderation
  - Mentoring



https://evergreen-ils.org/wp-content/uploads/2023/12/Annual%20Report%202022.pdf

Sustain: FOSS Projects

- Funding
  - Funding for the project costs
  - Funding for the core maintainers
  - Lifestyle funding
  - Funding for all contributors
  - Non-fiscal sponsorships



efile GRAPHIC print - DO NOT PROCESS | As Filed Data - | DLN: 93492133020512

Form **990EZ**

Department of the Treasury
Internal Revenue Service

Short Form
Return of Organization Exempt From Income Tax

Under section 501(c), 527, or 4947(a)(1) of the Internal Revenue Code (except private foundations)

▶ Do not enter social security numbers on this form as it may be made public.

▶ Go to www.irs.gov/Form990EZ for instructions and the latest information.

OMB No. 1545-0047

2021

Open to Public Inspection

A For the 2021 calendar year, or tax year beginning 01-01-2021 , and ending 12-31-2021

B Check if applicable:
☐ Address change
☐ Name change
☐ Initial return
☐ Final return/terminated
☐ Amended return
☐ Application pending

C Name of organization
EVERGREEN PROJECT

D Employer identification number
83-1064700

G Accounting Method: ☑ Cash ☐ Accrual Other (specify) ▶

H Check ▶ ☑ if the organization is **not** required to attach Schedule B (Form 990, 990-EZ, or 990-PF).

I Website: ▶https://evergreen-ils.org

J Tax-exempt status (check only one) - ☑ 501(c)(3) ☐ 501(c)( ) ◀ (insert no.) ☐ 4947(a)(1) or ☐ 527

K Form of organization: ☑ Corporation ☐ Trust ☐ Association ☐ Other

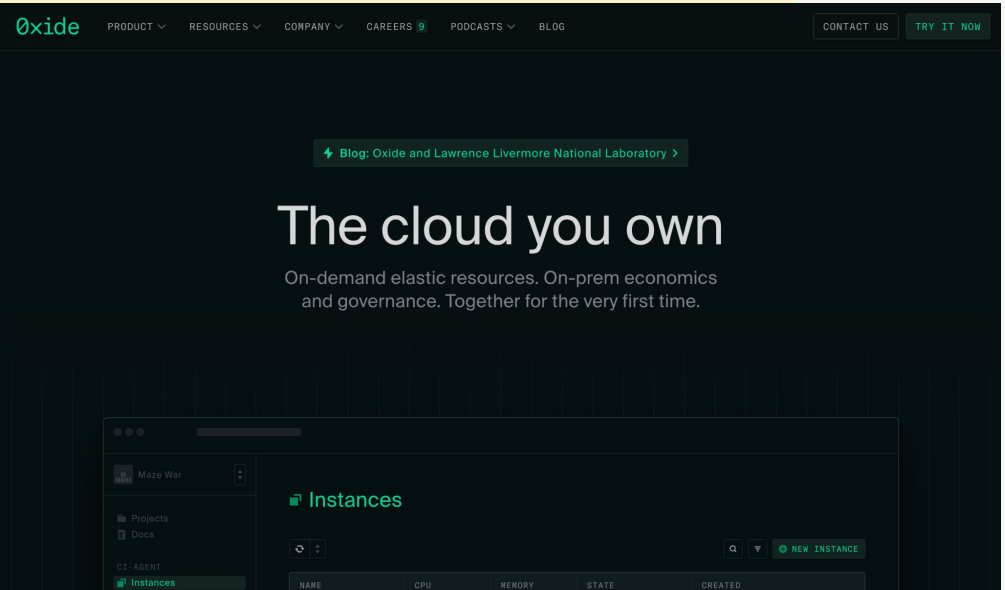
L Add lines 5b, 6c, and 7b to line 9 to determine gross receipts. If gross receipts are $200,000 or more, or if total assets (Part II, column (B) below) are $500,000 or more, file Form 990 instead of Form 990-EZ . . . . . ▶ $ 39,684

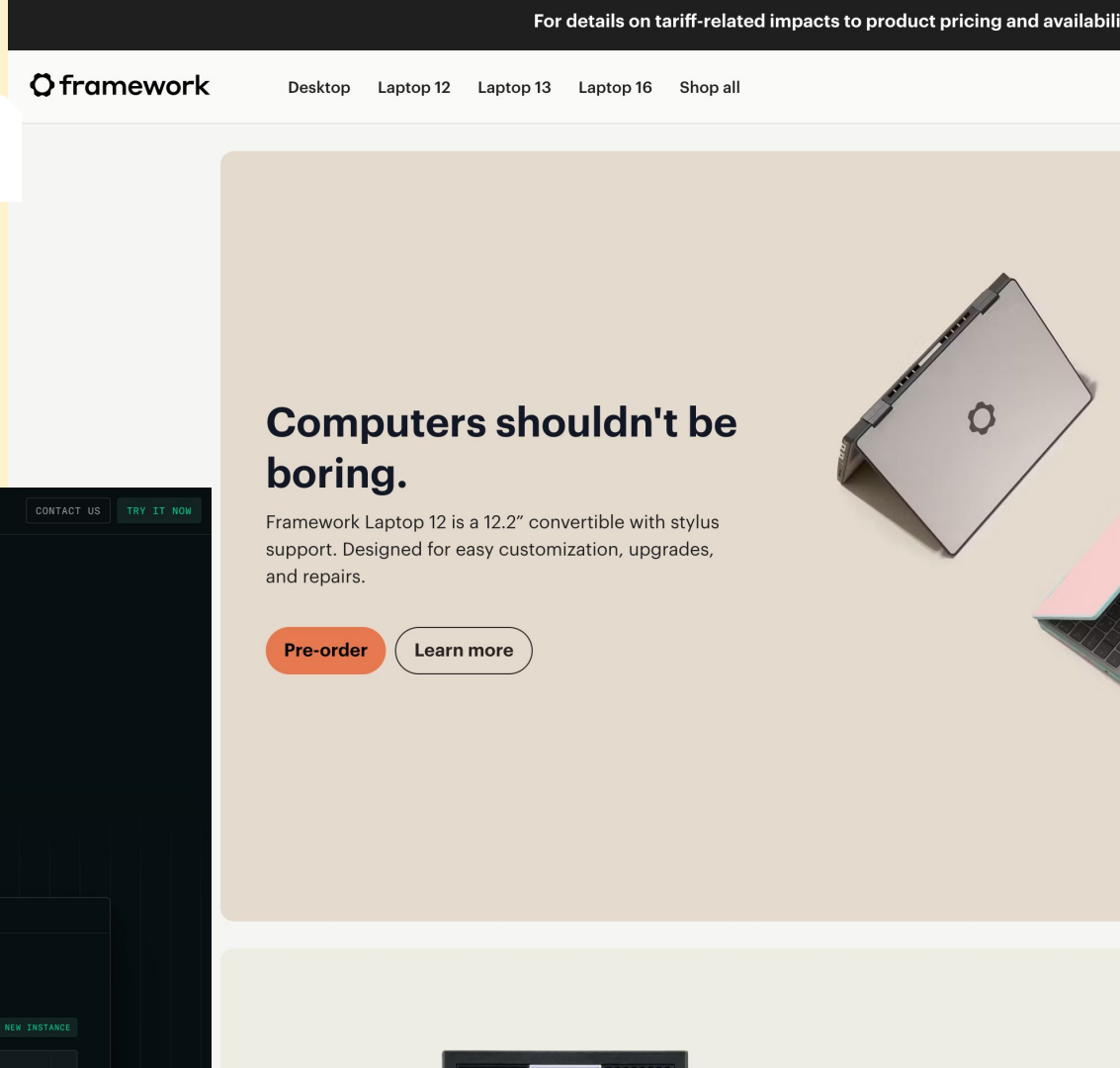
Part I **Revenue, Expenses, and Changes in Net Assets or Fund Balances** (see the instructions for Part I)
Check if the organization used Schedule O to respond to any question in this Part I . . . . . ☑

Revenue	1	Contributions, gifts, grants, and similar amounts received			1	
	2	Program service revenue including government fees and contracts			2	
	3	Membership dues and assessments			3	
	4	Investment income			4	
	5a	Gross amount from sale of assets other than inventory	5a	0		
	b	Less: cost or other basis and sales expenses	5b	0		
	c	Gain or (loss) from sale of assets other than inventory (Subtract line 5b from line 5a)			5c	
	6	Gaming and fundraising events				
	a	Gross income from gaming (attach Schedule G if greater than $15,000)	6a	0		
	b	Gross income from fundraising events (not including $ 0 of contributions from fundraising events reported on line 1) (attach Schedule G if the sum of such gross income and contributions exceeds $15,000)	6b	0		
	c	Less: direct expenses from gaming and fundraising events	6c	0		
	d	Net income or (loss) from gaming and fundraising events (add lines 6a and 6b and subtract line 6c)			6d	
	7a	Gross sales of inventory, less returns and allowances	7a	39		
	b	Less: cost of goods sold	7b	0		
	c	Gross profit or (loss) from sales of inventory (Subtract line 7b from line 7a)			7c	
	8	Other revenue (describe in Schedule O)			8	0

Sustain: FOSS Projects

- You can sell stuff:





Sustain: FOSS Developers

- Satisfaction
- Career trajectories
- Mentoring
- Funds and recognition
- Avoiding burnout



Sustain: FOSS Communities

- Generally the same, but at scale



Sustain: FOSS Businesses and Corporate

- Focus: a bottom line
- Not pushing OSS onto your employee's private time
- Having advocates and marketing inside the company for open source
- Risk
- OSPOs
- "Collaboration"

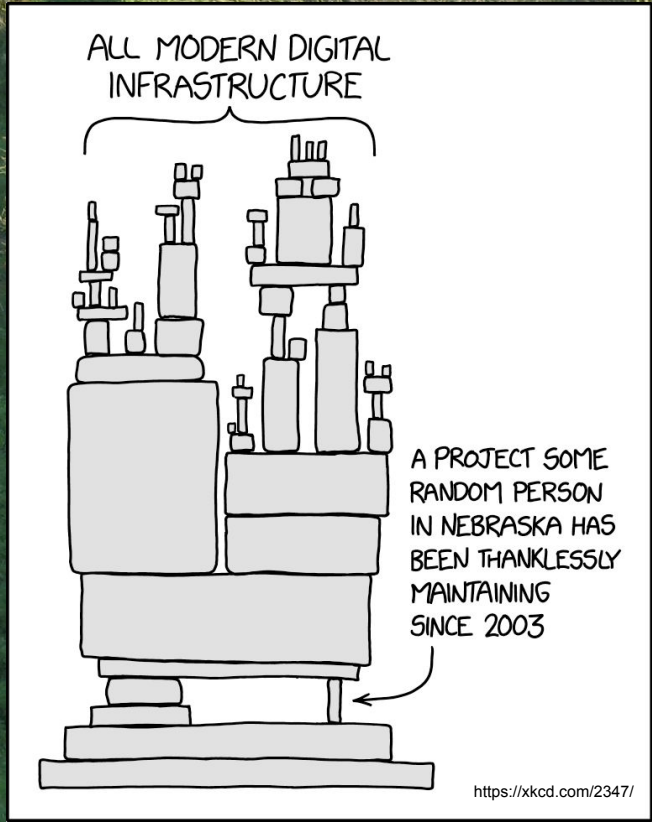


Sustain: FOSS Academia

- Not quite the same as industry or most open source
- Reproducibility
- Citations and altmetrics
- The project- or funding-based cycles of OSS in academia
- The student-graduation cycle of academia
- Where support comes from in the institution

Sustain: FOSS Ecosystem

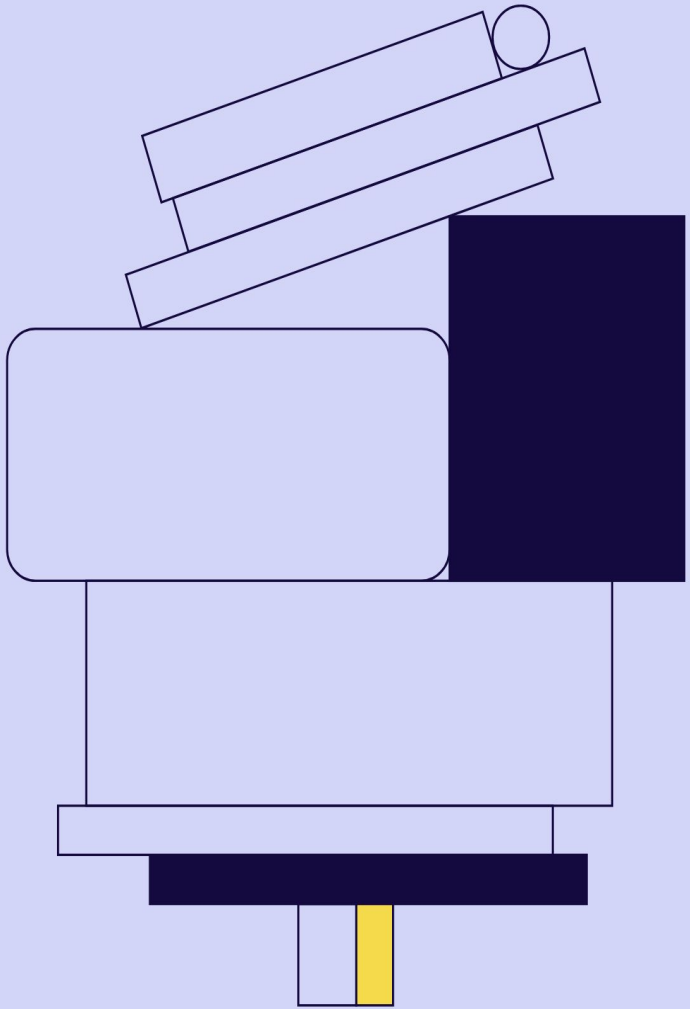
- Funding maintainers for their work
- Security and supply chains
- The "digital commons"



Gorse https://www.inaturalist.org/observations/65500592

Sustain: FOSS Ecosystem

- Legal sustainability and licenses
  - Stopping proliferation
  - Open washing
  - Patent issues
- AI + open source
- Taxation and infrastructural funding
- The CRA and their ilk



https://www.sovereigntechfund.de/

Sustain: FOSS Ecosystem

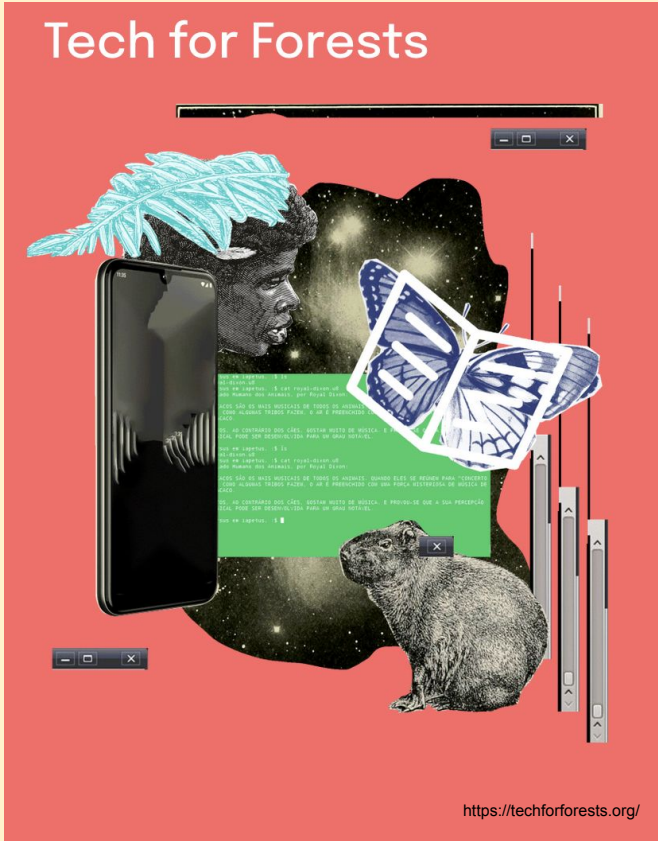


Sustain: FOSS & Environment

- Open source hardware
- Carbon-neutral data storage centers
- Reducing server costs
- Not using cryptocurrencies
- Avoiding the Conference trap

Sustain: FOSS & Environment

- Tech for environmental activists
- Tech for environmental studies
- Use of tech for the planet as a whole



Open Sustain Technology

- OpenSustain.tech
- ClimateTriage.com
- OpenClimate.fund
- OSS for Climate Podcast



- Climate Change
- Renewable Energy
- Energy Systems
- Atmosphere
- Biosphere
- Sustainable Development
- Hydrosphere
- Consumption
- Industrial Ecology
- Energy Storage
- Natural Resources
- Emissions
- Cryosphere


1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS
SUSTAINABLE DEVELOPMENT GOALS

UN OPEN SOURCE PRINCIPLES



Open by default
Making Open Source the standard approach for projects



Contribute back
Encouraging active participation in the Open Source ecosystem



Secure by design
Making security a priority in all software projects



Foster inclusive participation and community building
Encouraging active participation in the Open Source ecosystem



Design for reusability
Designing projects to be interoperable across platforms and ecosystems

Provide documentation
Providing thorough documentation for end-users, integrators and developers





RISE
(recognize, incentivize, support and empower)

Sustain and scale
Supporting the development



Linux Professional Institute

United Nations | open source week_

UN Headquarters, NYC | 16–20 June 2025





Open Research

- How can we build more collaborative practices in the sciences?
  - PLOS Computational Biology (in review): https://arxiv.org/abs/2505.06484
- How can we better map and show the value of OSS to the environmental movement?
  - OpenForum Aacademy Symposium (in prep): https://symposium.openforumeurope.org/
- What are "open practices"?
  - Nature Human Behavior (in review): https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4821969
- How can we use OSS tooling to protect our taonga manu?
  - My thesis work

Thank you

- https://burntfen.com
- http://richard.social

